

Caringbah High School

2015

Trial HSC Examination

Mathematics Extension I

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen (Black pen is preferred)
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–15, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2 – 4

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 5 – 10

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

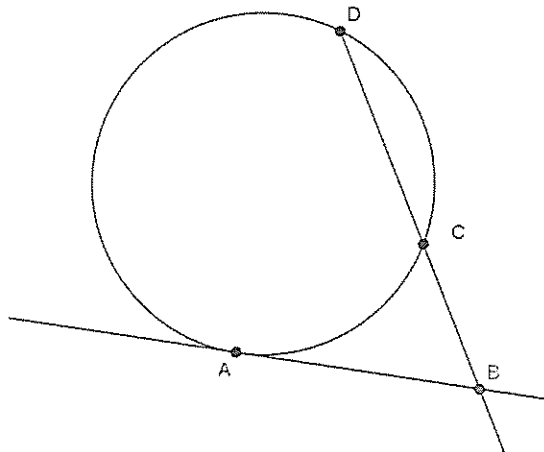
1) The exact value of $\tan \frac{\pi}{12}$ is:

- A) $\frac{1}{2\sqrt{3}}$ B) $(\sqrt{3}-1)^2$ C) $2+\sqrt{3}$ D) $2-\sqrt{3}$

2) The polynomial $p(x) = 2x^3 - x^2 - 6x + k$ has a factor $(x+2)$. What is the value of k ?

- A) 8 B) 0 C) -24 D) 32

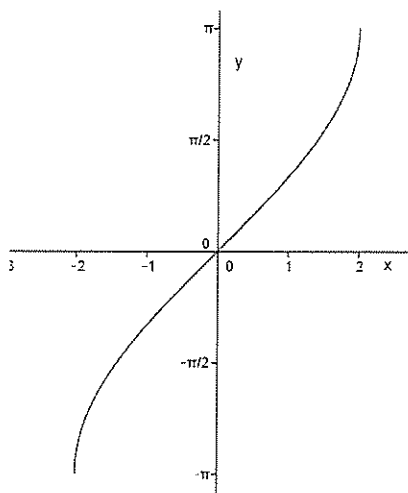
3) AB is a tangent at A



Which of the following is true?

- A) $AB = BC.BD$ B) $AB = BC.CD$
 C) $AB^2 = BC.CD$ D) $AB^2 = BC.BD$

- 4) The diagram shows the graph of a function. Which function does the graph represent?



- A) $y = 2 \cos^{-1}(2x)$ B) $y = 2 \sin^{-1}(2x)$
- C) $y = 2 \sin^{-1}\left(\frac{x}{2}\right)$ D) $y = 2 \cos^{-1}\left(\frac{x}{2}\right)$
- 5) Given the parametric equations $x = 2(\theta - \sin \theta)$ and $y = 2(1 - \cos \theta)$ which of the following represent $\frac{dy}{dx}$ in terms of θ ?

- A) $\frac{2 \sin \theta}{2 - \cos \theta}$ B) $\frac{1 - \cos \theta}{\sin \theta}$
- C) $\frac{\sin \theta}{1 - \cos \theta}$ D) $\frac{\sin \theta}{1 + \cos \theta}$

6) $\lim_{x \rightarrow \infty} \left[\frac{x+2}{1-x} \right] =$

- A) 1 B) -1 C) 2 D) -2

- 7) A particle is moving in simple harmonic motion and the acceleration is given by $\ddot{x} = -4x + 8$. The centre of the motion is:

A) -8 B) 8 C) 2 D) -2

- 8) Using the substitution $x = 1 - u^2$ then $\int \frac{x \, dx}{\sqrt{1-x}} =$

A) $-2 \int 1 - u^2 \, du$ B) $-2 \int u^2 - 1 \, du$

C) $\frac{1}{2} \int u^2 - 1 \, du$ D) $\frac{1}{2} \int 1 - u^2 \, du$

- 9) $A(1, -3)$ and $B(x, y)$ are 2 points, $P(-1, -1)$ divides these points A and B externally in the ratio $(2, 3)$. The co-ordinates of B are:

A) $(-2, -5)$ B) $(2, -4)$ C) $\left(\frac{2}{3}, \frac{-7}{3}\right)$ D) $(-2, 4)$

- 10) $y = f(x)$ is a linear function with slope $\frac{1}{3}$, the slope of $y = f^{-1}(x)$ is

A) 3 B) $\frac{1}{3}$

C) -3 D) $-\frac{1}{3}$

END OF MULTIPLE CHOICE QUESTIONS

Section II

60 marks

Attempt all questions 11–14 Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.

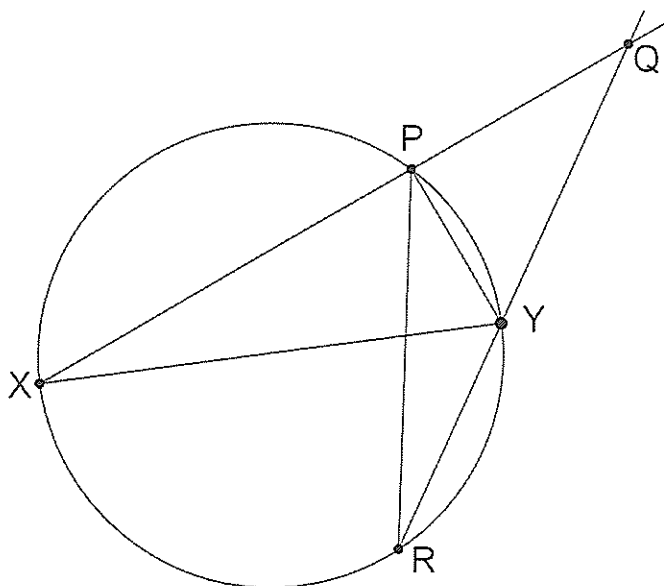
Marks

a) Find $\frac{d^2y}{dx^2}$ if $y = \ln(e^x + 1)$ 2

b) i) Show that $\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$ 2

ii) Hence find the value of $\tan 22\frac{1}{2}^\circ$ in simplest exact form 2

c)



XY is the diameter in the circle. 3

Given that $\angle PXY = 35^\circ$ and $\angle PQY = 25^\circ$,
Find the size of $\angle YPR$ giving reasons.

Question 11 continues on the next page

Question 11 continued.

Marks

d) $P(x) = x^3 + 3x^2 + 6x - 5$

i) Show that the equation $P(x) = 0$ has a root α such that $0 < \alpha < 1$ 2

ii) Use one application of Newton's method with a starting value of $x = 0.5$ to find an approximation for α . 2
Answer to 2 decimal places.

e) Find the exact value of 2

$$\int_{\frac{1}{6}}^{\frac{\sqrt{3}}{6}} \frac{3}{\sqrt{1-9x^2}} dx$$

End of Question 11

Question 12 (15 marks) Start a NEW booklet.**Marks**

a) Find $\lim_{x \rightarrow 0} \frac{\sin 2x + \tan x}{4x}$. 2

b) Use the substitution $u = \tan^{-1} x$ to evaluate the following. Leave your answer in exact form 3

$$\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$$

c) The function $f(x) = \frac{1}{1+3e^{-x}}$ is defined for all real x and $e^x > 0$

i) Sketch the curve $y = f(x)$ mark in any asymptotes, x, y intercepts 3

ii) Explain why an inverse function exists for $y = f(x)$ 1

iii) Find the inverse function $y = f^{-1}(x)$ 2

d) The volume, V of a spherical balloon of radius r mm is increasing at a constant rate of 400mm^3 per second.

i) Find $\frac{dr}{dt}$ in terms of r 2

ii) Find the rate of increase of the surface area S of the balloon when the radius is 25mm 2

End of Question 12.

Question 13 (15 marks) Start a NEW booklet.

Marks

- a) i) Sketch the graph of $y = 2 \cos^{-1} 2x$, show any intercepts with axes, and the domain and range. 2
- ii) The region in the first quadrant in the above graph is rotated about the y axis.
- α) Show that $x^2 = \frac{1}{4} \cos^2 \frac{y}{2}$ 1
- β) Find the volume of the solid formed (Answer in terms of π) 3
- b) Find $\int 2x^2 e^{4x^3+2} dx$ 2
- c) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$ where x is the displacement from the origin. Initially the object is at the origin with velocity (v) $2ms^{-1}$
- i) Prove that $V = 2e^{\frac{-x}{2}}$ 2
- ii) What happens to v as x increases without bound ? 1
- d) Use Mathematical Induction to show that $\cos(x + n\pi) = (-1)^n \cos x$ for all positive integers $n \geq 1$ 4

End of Question 13.

Question 14 (15 marks) Start a NEW booklet.**Marks**

- a) The acceleration $\ddot{x}$ m/s² at time, t seconds, of a particle moving in a straight line is given by

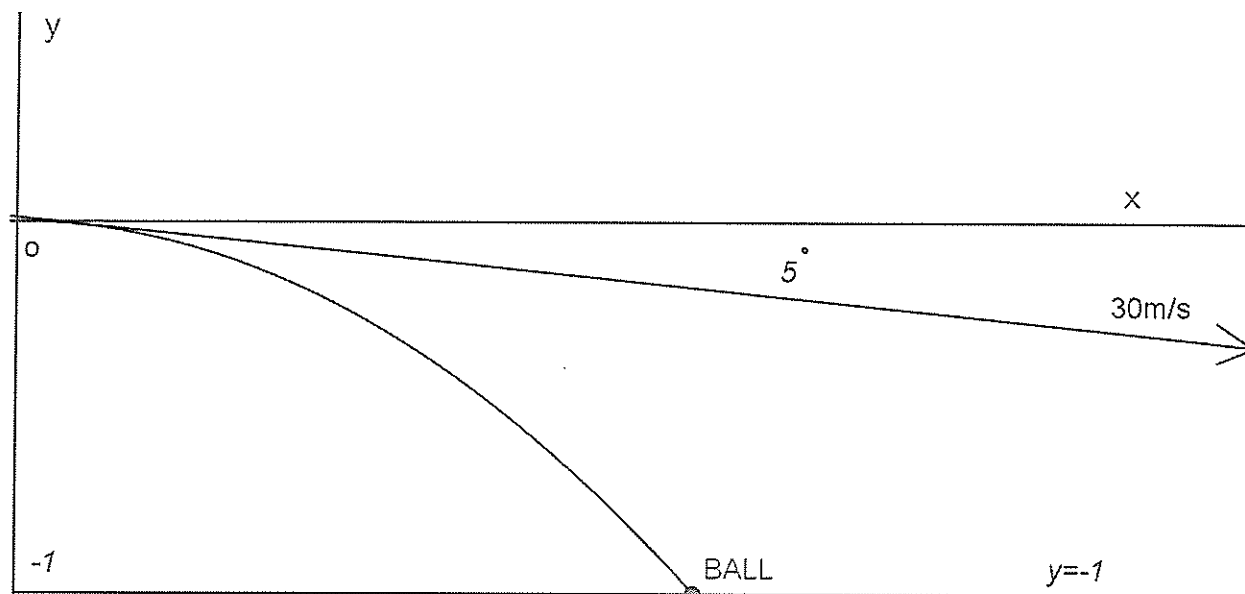
$$\ddot{x} = -4 \cos 2t - 8 \sin 2t$$

The particle is at a distance of x metres from the origin at time t and initially it is at $x = 1$ with a velocity of 4m/s

- | | | |
|------|---|---|
| i) | Show that $\ddot{x} = -4x$ | 3 |
| ii) | Show that the position of the particle after $\frac{\pi}{4}$ seconds is 2 metres to the right of the origin and the magnitude of its velocity is 2m/s at this time. | 2 |
| iii) | Is the speed of the particle increasing or decreasing when $t = \frac{\pi}{4}$.
Justify your answer. | 2 |

Question 14 continued.

Marks



b)

A batman hits a cricket ball which leaves the bat 1 metre above the ground with an initial speed of 30ms^{-1} at an angle of 5° in a downward direction. The equations of motion for the ball are $\ddot{x} = 0$ and $\ddot{y} = -10$

- i) Taking the origin to be the point where the ball leaves the bat, prove by using calculus that the ball has co-ordinates at time t given by 4

$$x = 30t \cos 5^\circ \quad \text{and}$$

$$y = -30t \sin 5^\circ - 5t^2$$

- ii) Find the time which elapses for the ball to strike the ground. (3dp) 2
- iii) Calculate the angle at which the ball strikes the ground. (nearest degree) 2

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Extension 1 Trial Exam Caringbah High School
Mathematics 2015

Multiple Choice

1, D 2, A 3, D 4, C 5, C 6, B 7, C 8, A 9, B 10, A

Question 11

a) $y = \ln(e^x + 1)$

$$y' = \frac{e^x}{e^x + 1}$$

$$y'' = \frac{(e^x + 1) \cdot e^x - e^x \cdot e^x}{(e^x + 1)^2}$$
$$= \frac{e^x}{(e^x + 1)^2}$$

b) (i) $\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$

b.h.s

$$\frac{1 - (\cos^2 x - \sin^2 x)}{1 + (\cos^2 x - \sin^2 x)}$$
$$= \frac{1 - (1 - 2\sin^2 x)}{1 + (2\cos^2 x - 1)}$$
$$= \frac{2\sin^2 x}{2\cos^2 x}$$
$$= \tan^2 x$$

ii) $\tan^2 22\frac{1}{2}^\circ = \frac{1 - \cos 45}{1 + \cos 45}$

$$\tan^2 22\frac{1}{2}^\circ = (\sqrt{2} - 1)^2$$

$$\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$$

c) Join PY and XR

$$\angle PXY = 35^\circ \quad \left. \begin{array}{l} \angle PQY = 25^\circ \\ \angle XRY = 90^\circ \end{array} \right\} \text{given}$$

$$\angle XRY = 90^\circ \quad (\text{angle in semicircle given } XY \text{ is diameter})$$

$$\angle YXR = 30^\circ \quad (\text{angle sum } \triangle QXR)$$

$$\angle YXR = \angle YPR = 30^\circ \quad (\text{angle at circ standing on same chord or arc})$$

Q11 cont'd

(i) $P(x) = x^3 + 3x^2 + 6x - 5$

(d) Since $P(0) = -5 < 0$ & $P(1) = 5 > 0$ and the curve is continuous, there is a root α between 0 and 1

(ii) $f(x) = x^3 + 3x^2 + 6x - 5$

$$f(0.5) = -1.125$$

$$f'(0.5) = 9.75$$

$$x_2 = 0.5 - \frac{(-1.125)}{9.75}$$

$$x_2 \approx 0.62 \quad (2 \text{ dp})$$

(e) $\int_{1/6}^{\sqrt{3}/6} \frac{3}{\sqrt{1-9x^2}} dx$

$$= 3 \int_{1/6}^{\sqrt{3}/6} \frac{1}{\sqrt{9(1/9 - x^2)}} dx$$
$$= \int_{1/6}^{\sqrt{3}/6} \frac{1}{\sqrt{(1/3)^2 - x^2}} dx$$

$$= \left[\sin^{-1} 3x \right]_{1/6}^{\sqrt{3}/6}$$
$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$$

$$= \pi/3 - \pi/6$$

$$= \pi/6$$

Question 12.

$$\begin{aligned}
 (a) \lim_{x \rightarrow 0} \frac{\sin 2x + \tan x}{4x} \\
 &= \frac{\sin 2x}{4x} + \frac{\tan x}{4x} \\
 &= \frac{1}{2} \cdot \frac{\sin 2x}{2x} + \frac{1}{4} \cdot \frac{\tan x}{x} \\
 &= \frac{1}{2} + \frac{1}{4} \\
 &= \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 b) \int_0^1 \frac{\tan^{-1} x}{1+x^2} dx & \quad \begin{array}{l} u = \tan^{-1} x \\ du = \frac{1}{1+x^2} dx \end{array} \\
 \int_0^{\pi/4} u \cdot du & \quad \begin{array}{l} x=1, u = \pi/4 \\ x=0, u = 0 \end{array} \\
 &= \left[\frac{u^2}{2} \right]_0^{\pi/4} \\
 &= \frac{1}{2} \left[\frac{\pi^2}{16} - 0 \right] \\
 &= \frac{\pi^2}{32}
 \end{aligned}$$

$$\begin{aligned}
 c) f(x) &= \frac{1}{1+3e^{-x}} \\
 &= \frac{1}{1+\frac{3}{e^x}}
 \end{aligned}$$

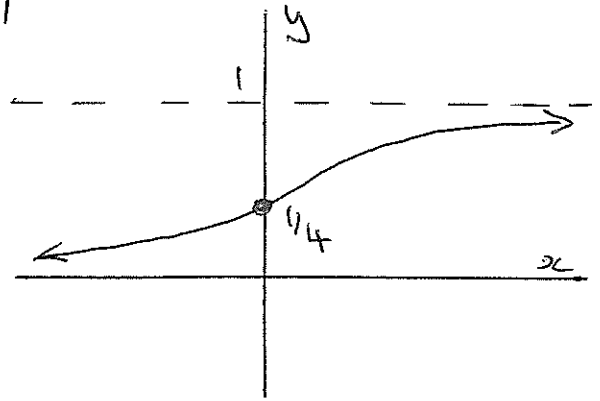
when $x=0$, $y = 1/4$

$x \rightarrow \infty$, $f(x) \rightarrow 1$

$x \rightarrow -\infty$, $f(x) \rightarrow 0$

(c) cont'd

(i)



(ii) By horizontal line test only 1 intercept

$$(iii) y = \frac{1}{1+3e^{-x}}$$

$$x = \frac{1}{1+3e^{-y}}$$

$$1+3e^{-y} = \frac{1}{x}$$

$$3e^{-y} = \frac{1}{x} - 1$$

$$\frac{3}{e^y} = \frac{1-x}{x}$$

$$\frac{e^y}{3} = \frac{x}{1-x}$$

$$e^y = \frac{3x}{1-x}$$

$$\ln e^y = \ln \left[\frac{3x}{1-x} \right]$$

$$y = \ln \left[\frac{3x}{1-x} \right]$$

$$d) \frac{dv}{dt} = 400 \text{ mm}^3/\text{s}, \quad V = \frac{4}{3}\pi r^3$$

$$(i) \frac{dv}{dt} = \frac{dv}{dr} \times \frac{dr}{dt}$$

$$400 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{100}{\pi r^2}$$

(ii)

$$\frac{ds}{dt} = \frac{ds}{dr} \times \frac{dr}{dt}$$

$$= 8\pi(25) \cdot \frac{100}{\pi(25)^2}$$

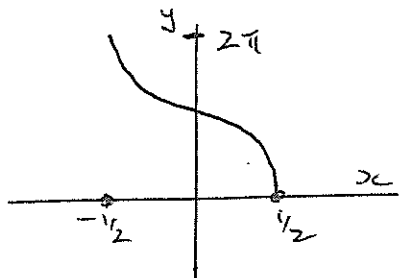
$$= 32 \text{ mm}^2/\text{s}$$

Question 13

(a) $y = 2 \cos^{-1} 2x$

i) $D: -1 \leq 2x \leq 1$
 $-\frac{1}{2} \leq x \leq \frac{1}{2}$

$R: 0 \leq y \leq 2\pi$



ii) $y = 2 \cos^{-1} 2x$

(a) $\frac{1}{2} \cos \frac{y}{2} = x$
 $x^2 = \frac{1}{4} \cos^2 \frac{y}{2}$

(b) $V = \pi \int_{\pi}^0 x^2 dy$

$V = \pi \int_0^{\pi} \frac{1}{4} \cos^2 \frac{y}{2} dy$

$V = \pi/4 \int_0^{\pi} \frac{1}{2} (1 + \cos y) dy$

$V = \pi/8 \left[y + \sin y \right]_0^{\pi}$

$V = \pi/8 \left[(\pi + \sin \pi) - 0 \right]$

$V = \frac{\pi^2}{8} u^3$

b) $\int 2x^2 e^{4x^3+2} dx$
 $= \frac{1}{6} \int 12x^2 e^{4x^3+2} dx$
 $= \frac{1}{6} e^{4x^3+2} + C$

(c) $\frac{d}{dx} x = -2e^{-2x}$

i) $\frac{d}{dx} \frac{1}{2} v^2 = -2e^{-2x}$

$\frac{1}{2} v^2 = 2e^{-2x} + C$

when $x = 0, v = 2$

$\frac{1}{2} (2)^2 = 2e^0 + C$

$2 = 2 + C$

$C = 0$

$\therefore \frac{1}{2} v^2 = 2e^{-2x}$

$v^2 = 4e^{-2x}$

$v = (4e^{-2x})^{1/2}$

$v = 2e^{-x/2}$

ii) $x \rightarrow \infty$

$v = \frac{2}{e^{x/2}}$

$e^{x/2} \rightarrow \infty \therefore v \rightarrow 0$

(d) $\cos(x + n\pi) = (-1)^n \cos x, n \geq 1$

Prove true for $n=1$

$\cos(x + \pi) = (-1)^1 \cos x$

$-\cos x = -\cos x$

Assume true for $n=k$

$\cos(x + k\pi) = (-1)^k \cos x$

Prove true for $n=k+1$

L.H.S. $\cos[x + (k+1)\pi]$

$= \cos[(x + k\pi) + \pi]$

$= \cos(x + k\pi) \cos \pi - \sin(x + k\pi) \sin \pi$

$= -1 [\cos(x + k\pi)]$

$= -1 [(-1)^k \cos x]$

$= (-1)^{k+1} \cos x$

plus M.I. statement

Question 14

$$i) \ddot{x} = -4 \cos 2t - 8 \sin 2t$$

$$ii) \dot{x} = -2 \sin 2t + 4 \cos 2t + c$$

$$\text{when } t=0, \dot{x}=4 \therefore c=0$$

$$\dot{x} = -2 \sin 2t + 4 \cos 2t$$

$$x = \cos 2t + 2 \sin 2t + c$$

$$t=0, x=1, \therefore c=0$$

$$\therefore x = \cos 2t + 2 \sin 2t$$

$$= 4x = -4 \cos 2t - 8 \sin 2t$$

$$\therefore \ddot{x} = -4x$$

$$(ii) x = \cos 2t + 2 \sin 2t$$

$$\text{when } t = \pi/4$$

$$x = \cos \pi/2 + 2 \sin \pi/2$$

$$x = 0 + 2(1)$$

$$x = 2$$

$$\text{when } t = \pi/4$$

$$\dot{x} = -2 \sin \pi/2 + 4 \cos \pi/2$$

$$\dot{x} = -2(1) + 4(0)$$

$$\dot{x} = -2$$

$$(iii) \text{ when } t = \pi/4, x=2, \dot{x} = -2 \text{ m/s}$$

$$\ddot{x} = -4x$$

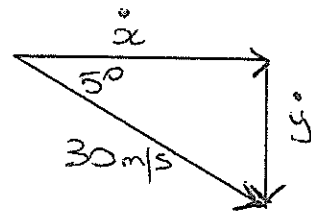
$$= -4(2)$$

$$= -8 \text{ m/s}^2$$

$\therefore$ Speed increasing

as $\ddot{x} < 0, \dot{x} < 0$

b) Initial velocity diagram



$$\text{Vert } \sin 5^\circ = \frac{\dot{y}}{30}$$

$$-30 \sin 5^\circ = \dot{y}$$

$$\text{Hor } \cos 5^\circ = \frac{\dot{x}}{30}$$

$$30 \cos 5^\circ = \dot{x}$$

Equations of motion

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$x = 30t \cos 5^\circ$$

$$x = 30t \cos 5^\circ + c$$

$$t=0, x=0, c=0$$

$$x = 30t \cos 5^\circ$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$t=0, \dot{y} = -30 \sin 5^\circ$$

$$\therefore c = -30 \sin 5^\circ$$

$$\dot{y} = -10t - 30 \sin 5^\circ$$

$$y = -5t^2 - 30t \sin 5^\circ + c$$

$$t=0, y=0, c=0$$

$$y = -5t^2 - 30t \sin 5^\circ$$

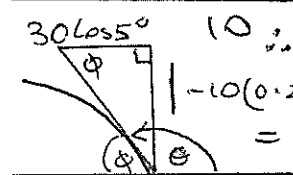
$$(i) -1 = -5t^2 - 30t \sin 5^\circ$$

$$5t^2 + 30t \sin 5^\circ - 1 = 0$$

$$t = \frac{-30 \sin 5^\circ \pm \sqrt{(30 \sin 5^\circ)^2 + 20}}{10}$$

$$30 \cos 5^\circ \quad 10 \therefore t = 0.2566$$

$$(ii) -10(0.2566) - 30 \sin 5^\circ = 5.18067228$$



$$\tan \phi = \frac{5.18067228}{30 \cos 5^\circ}$$

$$\phi = 9^\circ 50' 3.91''$$

Ball strikes ground at

$$\theta = 170^\circ 9' 56.09''$$